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ANTIHYPERTENSIVE AGENTS: MECHANISMS OF DRUG ACTION (ME ERNST, SECTION EDITOR)

Drug Interactions with Antihypertensives

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Abstract

Purpose of Review Hypertension is remarkably prevalent, affecting an estimated 1.13 billion people worldwide. It often requires the use of multi-drug regimens and is commonly associated with a myriad of other comorbidities which increase medication use. The pervasive use of antihypertensive medications combined with the presence of polypharmacy in many hypertensive patients results in significant risk of drug interactions. This review will summarize the relevant literature to assist clinicians in mitigating drug interaction risks when prescribing antihypertensives.

Recent Findings Pharmacokinetic interactions affect drug disposition in the body and can occur at the steps of absorption, distribution, metabolism, or elimination of involved medications. Data has established the calcium channel blockers, namely, diltiazem and verapamil, as potent inhibitors of CYP3A4, and the majority of significant drug interactions involving antihypertensives are attributable to these two agents. Although less common, pharmacokinetic drug interactions with other antihypertensive classes have also been identified. Pharmacodynamic drug interactions with antihypertensives lead to synergy or antagonism of blood pressure lowering effects and can increase or mitigate adverse effects depending on the agents involved. Knowledge is emerging about drug-induced phenoconversion, a phenomenon whereby a drug interaction results in a drug metabolizing phenotype that is different than that predicted by an individual's genotype.

Summary Antihypertensive use in patients with comorbidities and polypharmacy increases the likelihood of encountering important drug-drug interactions. Dedicated efforts to better understand the relationship between pharmacokinetic drug interactions and pharmacogenomic information is important to advance efforts related to personalized medicine.

Keywords Drug-drug interactions · Antihypertensive agents · Pharmacokinetic drug interactions · Pharmacodynamic drug interactions

Introduction

According to the World Health Organization, an estimated 1.13 billion people worldwide have hypertension, with less than one in five achieving optimal blood pressure control [1]. Almost half of the US population, 108 million people, meet criteria for the diagnosis of hypertension, with 81 million qualifying to receive antihypertensive treatment according to most recent hypertension guidelines [2]. Patients with hypertension often have multiple comorbidities necessitating diverse pharmacotherapies and leading to high medication burden. For example, in the Systolic Blood Pressure Intervention Trial (SPRINT), 40% of the 9361 patients with hypertension included in the trial were taking five or more medications at baseline [3].

While the clinical impact of drug interactions with antihypertensives may be less prominent or severe than the classic drug interactions involving, for example, antiepileptic or antiinfective medications, the incidence of drug interactions occurring with antihypertensive drugs is substantial. One study, a longitudinal analysis of antihypertensive drug use in a Medicaid population, found an interaction in 75% of the patients evaluated, with 35% of those classified as highly significant [4••]. The often subtle nature of these interactions should not diminish their significance, and familiarity with the basic

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principles underlying drug interactions is key to managing their risk when prescribing and monitoring antihypertensives drugs. The purpose of this review is to provide an overview of antihypertensive drug-drug interactions to assist clinicians in minimizing their adverse impact.

Drug Interaction Overview

A drug interaction can manifest in two basic ways but ultimately occurs when one drug impacts the pharmacologic, therapeutic, or adverse event profile of another co-administered drug. Drug interactions can be broadly classified into two major categories, pharmacokinetic and pharmacodynamic. Pharmacokinetic drug interactions occur when one drug (often referred to as the precipitant drug) modifies the absorption, metabolism, distribution, or elimination of another co-administered drug (often referred to as the substrate drug). Pharmacodynamic drug interactions occur when a drug alters the end-pharmacologic effect of another drug (i.e., the therapeutic efficacy or adverse effect profile) without necessarily altering the drug's disposition in the body. While, in general, attempts should be made to minimize or avoid drug interactions if possible, there are selected situations when pharmacodynamic interactions with antihypertensives can be both beneficial and desirable.

Pharmacokinetic Drug Interactions

Absorption

Antihypertensive medications as a therapeutic class are, for the most part, well absorbed with oral bioavailability profiles that are relatively unaffected by other medications. However, there are certain co-prescribed medications which may interfere with absorption. For example, newer potassium binders have demonstrated efficacy in the treatment of hyperkalemia specifically in patients using renin-angiotensin-aldosterone system (RAAS) inhibitors. Patiromer, one agent in this class, has demonstrated reduced in vivo serum concentration of amlodipine, furosemide, and metoprolol, likely through gastrointestinal binding and excretion. Although these specific antihypertensive agents are not RAAS inhibitors, they are often co-prescribed in patients receiving a RAAS inhibitor and may likely be used in combination with patiromer in clinical practice. To avoid reduced medication absorption and the potential for decreased therapeutic effects, the timing of medication administration is critical. Package labeling suggests 3 h of separation between patiromer and all other oral medications, an administration guideline important to adhere to when co-prescribing with antihypertensives [5]. Interestingly, sodium zirconium cyclosilicate, another potassium binder, alters gastric pH and has an opposite effect in that it can lead to increased absorption of drugs like furosemide which have pH-dependent absorption [6]. Given that both furosemide and sodium zirconium cyclosilicate can reduce potassium levels, the increased risk of hypokalemia with this combination is an important interaction to consider clinically.

Although rarely used now for treatment of dyslipidemia, bile acid sequestrants exert their cholesterol lowering effects via an anion exchange resin which may also bind drugs administered in close proximity and lead to their diminished absorption. Cholestyramine has been associated with decreased absorption of propranolol and thiazide diuretics [7]. Similar to patiromer, bile acid sequestrants should be administered as far apart from other medications as possible to minimize this possible interference.

While it is not commonly used in the general population, methyldopa remains commonly used for the treatment of hypertension in pregnant women. A clinically significant interaction between oral iron and methyldopa has been reported, resulting in up to 50% decreased absorption of methyldopa, likely due to chelating effects of iron [8]. Given that iron is a commonly used supplement in pregnancy, this is a potentially important interaction which could reduce the antihypertensive efficacy of methyldopa. The methyldopa dose can likely be titrated to account for the decreased absorption, but iron nonadherence or discontinuation due to intolerance may alter this balance and lead to an unrecognized cause of blood pressure fluctuation and potentially further unnecessary work-up and treatment.

Metabolism

Cytochrome P-450 (CYP) enzymes are responsible for the metabolism of numerous medications. In addition to being substrates undergoing metabolism by these enzymes, drugs can also induce or inhibit their action, which leads to the potential for drug interactions. With regard to antihypertensive medications, there are a few key instances of drug interactions resulting from altered metabolism which have meaningful clinical implications.

CYP2D6, one specific CYP enzyme subtype, is involved in the metabolism of several beta-blockers including bisoprolol, carvedilol, metoprolol, labetalol, nadolol, and propranolol [9]. When co-administered with medications that induce CYP2D6 activity, beta-blockers can have decreased serum concentrations which potentially reduces their therapeutic efficacy. Inhibitors of CYP2D6, on the other hand, may result in increased serum concentration of these beta-blockers and potentially increased adverse effects when co-administered. CYP2C9 is another prominent drug metabolizing enzyme and in the case of antihypertensives is involved in metabolism of two angiotensin receptor blockers (ARBs), irbesartan and losartan, and the loop diuretic, torsemide [9, 10]. Similar impact on serum concentration of these antihypertensives as described previously for CYP2D6 would be expected when

exposed to inhibitor/inducers of this particular enzyme. Table 1 lists selected medications with known impact on CYP2D6 and CYP2C9 function and antihypertensive agents that are metabolized primarily via this pathway; however, it is important to recognize that the clinical significance of cytochrome P450–based drug interactions may vary greatly based on the level of contribution the particular enzyme has on the overall metabolism and, ultimately, removal and/or deactivation of a particular drug.

CYP3A4 is involved in the metabolism of calcium channel blockers (i.e., diltiazem, verapamil, amlodipine, felodipine, nicardipine, and nifedipine) and one ARB, losartan [9, 10]. When inhibitors or inducers of CYP3A4 are co-prescribed with these medications, serum concentration of these antihypertensives can be affected similarly to that described in the previous examples for CYP2D6 and CYP2C9. More importantly, verapamil and diltiazem, and to a lesser extent amlodipine and nifedipine, can also act as inhibitors themselves of CYP3A4 [9]. It should be noted that the potency with which medications inhibit or induce metabolizing enzymes varies widely, with more potent effects on enzyme activity leading to greater clinical consequences [11]. This leads to the possibility of clinically significant drug interactions when antihypertensives with potent CYP3A4 inhibiting effects, like diltiazem and verapamil, are co-prescribed with drugs that rely on this enzyme for metabolism. This is evident even within the calcium channel blocker class, with clinical trial evidence demonstrating increased serum concentration of nifedipine (a CYP3A4 substrate) when taken concurrently with verapamil or diltiazem (potent CYP3A4 inhibitors) [14].

Three HMG-CoA reductase inhibitors (e.g., "statins"), simvastatin, atorvastatin, and fluvastatin, undergo metabolism via CYP3A4 and are commonly co-prescribed in patients with hypertension [15]. Inhibition of this enzyme via verapamil and diltiazem or, to a lesser extent, amlodipine can lead to increased statin serum concentration and enhanced risk for myopathy. As a result of this interaction, the FDA has provided dosing guidelines for simvastatin when used in combination with one of these identified calcium channel blockers [16]. Atorvastatin and fluvastatin are less impacted by inhibition of this metabolism, and dosing adjustment recommendations have not been established for either agent; nevertheless, the potential for this interaction should be considered when co-prescribing these agents with calcium channel blockers. Anticoagulants are another class of medication commonly co-prescribed with antihypertensive medications. The direct oral anticoagulants offer an advantage over warfarin with fewer drug interactions; however, their potential for interaction exists and should not be overlooked simply given less frequency of occurrence. Rivaroxaban undergoes significant metabolism via CYP3A4 and, therefore, may warrant dose adjustment and/or added monitoring when administered with diltiazem or verapamil. This may be particularly important in the setting of renal impairment given the significant contribution of renal elimination to the overall pharmacokinetic profile of rivaroxaban [17•]. Gout, another comorbidity widely associated with hypertension, is often treated with colchicine, which undergoes metabolism via CYP3A4. Given the serious side effects associated with colchicine toxicity, formal dosage adjustments are recommended in the FDA-approved

Table 1 Antihypertensive drugs and their disposition in the cytochrome P450 enzyme system and P-glycoprotein pathway [9, 11–13]

CYP3A4		CYP2D6		CYP2C9		P-Glycoprotein	
Inhibitors	Inducers	Inhibitors	Inducers	Inhibitors	Inducers	Inhibitors	Inducers
Clarithromycin	Carbamazepine	Amiodarone	None	Amiodarone	Carbamazepine	Amiodarone	Carbamazepine
Diltiazem*	Phenobarbital	Cimetidine		Fluconazole	Phenobarbital	Carvedilol*	Phenytoin
Erythromycin	Phenytoin	Diphenhydramine		Fluoxetine	Phenytoin	Clarithromycin	St. John's Wort
Itraconazole	St. John's Wort	Fluoxetine		Metronidazole	Rifampin	Itraconazole	Rifampin
Ketoconazole	Rifampin	Paroxetine		Trimethoprim/sulfamethoxazole		Ranolazine	
Verapamil*		Terbinafine				Verapamil*	
Antihypertensive substrates of CYP3A4		Antihypertensive substrates of CYP2D6		Antihypertensive substrates of CYP2C9		Antihypertensive substrates of P-glycoprotein	
ARBs		Beta-blockers		ARBs		Alpha-adrenergic blockers	
Losartan		Bisoprolol		Irbesartan		Doxazosin	
Calcium channel blockers		Carvedilol		Losartan		Beta-blockers	
Amlodipine		Labetalol		Diuretics		Nadolol	
Felodipine		Metoprolol		Torsemide		Calcium channel blockers	
Nifedipine		Nebivolol				Verapamil	
Diltiazem		Propranolol					
Verapamil							

This table includes a listing of common inhibitors and inducers of three primary cytochrome P450 enzyme systems (CYP3A4, CYP2D6, and CYP2C9) and the P-glycoprotein pathway (top row) and antihypertensive agents which are substrates of these systems (bottom row). Note that select antihypertensive agents have inhibiting properties as denoted by an asterisk

labeling information when co-administered with verapamil and diltiazem. These include an approximately 50% reduction in the original intended dose. For example, colchicine dosing for prophylaxis of gout when diltiazem or verapamil are co-prescribed is recommended to be 0.3 mg once daily instead of 0.6 mg daily, and treatment of an acute gout attack is recommended using 1.2 mg × 1 dose compared to 1.2 mg × 1 dose followed by 0.6 mg 1 h later [18].

The previous examples highlight the important clinical implications of pharmacokinetic drug interactions occurring at the metabolism step with antihypertensive medications. While there may be subtle impact of other medications on the serum concentration of antihypertensive agents which can increase and/or decrease their antihypertensive effects, these impacts can usually be monitored for, and doses adjusted, as part of routine assessment, posing little risk for serious harm to the patient. On the other hand, the impact that calcium channel blockers, specifically diltiazem and verapamil, have on the metabolism of other drugs given their inhibition of the CYP3A4 enzyme requires perhaps more attention, as the drugs impacted may have a narrow therapeutic index and high-risk toxicities associated with a supratherapeutic serum concentration. Clinicians should be cognizant of the potential for significant drug interactions when prescribing pharmacotherapy in patients taking verapamil or diltiazem. Consideration of alterative agents with less risk for drug interactions may be necessary, and if combining these antihypertensives with interacting medications is unavoidable, careful monitoring for adverse effects is necessary.

Excretion

Management of hypertension in patients taking chronic lithium therapy poses a unique challenge in many ways. Lithium itself causes development of chronic kidney disease over time, another potential contributor to blood pressure elevation. For this reason, concomitant use of lithium and antihypertensive therapy may be commonly encountered among patients with bipolar disease.

Lithium is a medication with a very narrow therapeutic index, where slight variations in blood concentration can have relevant clinical implications. Elevated serum concentration may produce lithium toxicity, and decreased concentration can lead to undertreated bipolar disease, both resulting in the potential for serious harm to the patient. Several classes of antihypertensives have been demonstrated to have significant impact on the excretion of lithium and result in altered lithium concentrations. Lithium undergoes reabsorption in the renal proximal and distal tubules via the sodium-hydrogen exchanger channel (proximal tubule) and the epithelial sodium channel (distal tubule). By decreasing aldosterone levels, RAAS inhibitors (angiotensin-converting enzyme inhibitors [ACEIs], ARBs, aldosterone antagonists) lead to decreased reabsorption of sodium in the distal tubule and through their action to increase potassium levels cause decreased sodium reabsorption in the proximal tubule. This decrease in sodium reabsorption in both distal and proximal tubules leads to a compensatory increase in sodium reabsorption via the sodium hydrogen exchanger and epithelia sodium channel, thereby increasing lithium reabsorption and serum concentration [19]. Similarly, thiazide diuretics inhibit sodium reabsorption in the distal tubule and induce natriuresis. This results in a compensatory increase in sodium reabsorption in the proximal tubule, which can then lead to a similar increase in lithium reabsorption at this site. Clearance of lithium has been projected to be reduced by 25–40% when used with a thiazide and ~25% with concomitant ACEI therapy [20]. It is generally best to avoid both thiazide diuretics and RAAS inhibitors when treating hypertension in patients receiving chronic lithium therapy.

P-glycoprotein

P-glycoprotein (P-gp) is a plasma membrane protein with actions that span across all pharmacokinetic parameters, including absorption, metabolism, excretion, and distribution of drugs. P-gp can be broadly thought of as having efflux properties; however, the specific function varies based on the particular location of the P-gp, as these proteins are found all throughout the body. In the intestinal lumen, a primary role of P-gp is to prevent drug absorption in the gut. In the brain, P-gp aids in the prevention of transit across the blood-brain barrier. In the kidneys, P-gp augments drug excretion. As such, inhibition of P-gp can lead to enhanced serum concentration of drugs which are substrates of P-gp, whereas induction of P-gp can lower concentrations of these drugs. It should be noted that P-gp activity is very complex, and exclusions and variations from these typical activity patterns can be seen. One example of this complexity is the fact that P-gp and CYP-3A4 enzymes are often found in the same organs and there is significant overlap in drugs that interact with both CYP-3A4 and P-gp. This can complicate the ability to quantify the clinical impact of P-gp alone versus its influence on CYP-3A4 actions on drugs [21].

One example of a clinically relevant drug interaction involving antihypertensives and P-gp is that between verapamil and dabigatran. Dabigatran is a P-gp substrate, and verapamil has known P-gp inhibitor properties. Although the impact that verapamil has in inhibiting P-gp efflux of dabigatran is not significant enough to warrant dabigatran dosing reductions in the FDA labeling, it is recommended that dabigatran dose reduction be considered in patients with reduced renal function who are using concomitant verapamil [17, 22]. Digoxin is another example of a cardiovascular medication that is a substrate of P-gp. Carvedilol and verapamil have been shown to result in inhibition of P-gp efflux of digoxin leading to increased blood concentration of digoxin [23]. Given that

verapamil, dabigatran, and digoxin are used to manage conditions that are related and may be encountered within the same patient, these are important interactions for clinicians to be aware of when prescribing in this population [24, 25].

Many drugs that inhibit P-gp are substrates of P-gp themselves, and verapamil is no excpetion [25]. Erythromycin inhibits P-gp, thereby increasing verapamil serum concentration, and St. John's Wort, an inducer of P-gp, has the opposite impact on verapamil concentration [9]. The clinical relevance of altered verapamil serum concentration may seem less important than that of other medications with a narrow therapeutic index. However, it is important to consider the entire scale of impact, keeping in mind the effect that verapamil has on the pharmacokinetics of other medications and how shifts in verapamil serum concentration may result in downstream effects on the metabolism of other medications.

Table 2 summarizes the key clinical pharmacokinetic drug interactions affecting antihypertensives.

Drug-Induced Phenoconversion

Patients can be classified as poor vs. extensive metabolizers for specific CYP metabolizing enzymes based on their genotype, which may have consequences for prescribing certain drugs. Despite advances in the availability of genetic testing and ability to interpret the information, it is not uncommon for genetic predictions to fail to manifest in clinical results. For example, patients predicted to have poor metabolism of a particular drug and who would be identified as candidates for reduced drug dosing may not end up needing this reduced dose as predicted. While there are many potential explanations for this, one common reason in patients on multi-drug therapies may be the phenomenon of drug-induced phenoconversion. Phenoconversion occurs when an individual's genotype-predicted drug metabolizing phenotype is transformed into another by external factors which can include co-administration of a drug which alters the pharmacokinetics of the substrate drug. For example, a patient with genotype-predicted poor metabolizer status for a substrate drug who is subsequently prescribed a drug which induces metabolism of the substrate drug may then have normal or extensive metabolism of the substrate drug due to the drug interaction. They are said to have "phenoconverted" to the normal or extensive metabolizer phenotype despite possessing a genotype which would have predicted poor metabolizer status. To date, there has been little research into how antihypertensives may be affected by this and no specific dosing recommendations exist; however, this is an example of how the evolving field of pharmacogenomics combined with knowledge of pharmacokinetics may be used in the future to achieve more personalized medicine [26•, 27•].

Pharmacodynamic Interactions

In contrast to pharmacokinetic drug interactions which affect a drug's disposition within body systems, pharmacodynamic interactions do not result in altered processing or handling of medications throughout the body. Instead, pharmacodynamic drug interactions lead to altered end organ physiologic action, either through synergistic, additive, or antagonistic effects. Pharmacodynamic drug interactions may be considered more easily identifiable given that they occur based on known mechanisms and actions of drugs. Definitions of synergistic, additive, and antagonistic effects generally refer to more than the anticipated effect of each drug individually (synergistic), the sum of individual drug effects (additive), and diminished effect of one drug due to opposing effects of the other drug (antagonistic).

Using combinations of blood pressure medications from different classes is a strategy often required in the majority of patients with hypertension. The additive blood pressure lowering effects of multiple antihypertensives is required in approximately two-thirds of patients to achieve guideline targets and, ultimately, reduce end organ damage [28]. Thoughtful selection of antihypertensive combinations is often based on predicted pharmacodynamic interactions. For example, combining an ACEI, ARB, or aldosterone antagonist with a thiazide diuretic can diminish the hypokalemic effect anticipated with the thiazide. Furthermore, diuretics have demonstrated an ability to activate the RAAS; therefore, this same combination can augment the blood pressure lowering ability of each medication class when used together.

Avoiding or minimizing use of certain combinations is a key consideration to evade undesirable pharmacodynamic effects. Common examples include the added negative inotropic effects when combining a beta-blocker with a non-dihydropyridine calcium channel blocker, or the compounded hyperkalemic effects of combining two RAAS inhibitors or a RAAS inhibitor with sulfamethoxazole/trimethoprim, or a non-steroidal anti-inflammatory drug (NSAID), especially when renal function is impaired. Other less evident interactions exist, including that between non-cardio-selective beta-blockers without alpha-blocking properties (e.g., propranolol) and clonidine, where there is potential for unopposed alpha agonism, and resultant hypertensive crisis, with beta-blocker withdrawal [29].

Outside of blood pressure medication combinations, there is the potential for a multitude of pharmacodynamic drug interactions between antihypertensives and non-antihypertensive medications. Examples of desirable pharmacodynamic interactions range from very global interactions, like that between antihypertensives and other cardiovascular risk lowering medications (e.g.., statins, sodium-glucose transport protein-2 [SGLT-2] inhibitors) which can have additive effects to improve

Table 2 Pharmacokinetic drug interactions with antihypertensives

Substrate drug	Precipitant drug	Clinical effect
Absorption		
Thiazide diuretics, propranolol	Cholestyramine	Decreased absorption of thiazide leading to reduced clinical effects (e.g., reduced blood pressure lowering) Decreased absorption of beta-blocker leading to reduced clinical effects (e.g., reduced blood pressure lowering, minimized bradycardic effects in atrial fibrillation, less ability to reduce afterload in heart failure)
Amlodipine, furosemide, metoprolol	Potassium binder (patiromer)	Decreased absorption of antihypertensives; effect diminished with separation of administration by 3 h
Furosemide	Potassium binder (sodium zirconium cyclosilicate)	Increased absorption of furosemide due to increased gastric pH leading to increased clinical effects (e.g., increased risk of hypokalemia); effect diminished with separation of administration by 2 h
Methyldopa	Iron salts	Decreased absorption of methyldopa leading to reduced clinical effects (e.g., reduced blood pressure lowering)
Metabolism		
Beta-blockers Bisoprolol Carvedilol Metoprolol Propranolol	CYP-2D6 inhibitors	Increased beta-blocker concentration leading to enhanced clinical effects (e.g., hypotension and bradycardia)
Calcium channel blockers (CCBs) Verapamil Diltiazem	CYP-3A4 inhibitors	Increased CCB concentration leading to enhanced clinical effects (e.g., hypotension and bradycardia)
	CYP-3A4 inducers	Decreased CCB concentration leading to reduced clinical effects (e.g., minimization of blood pressure and pulse lowering)
Calcineurin inhibitors (CNI) Tacrolimus Cyclosporine	CCBs (diltiazem and verapamil) -via CYP-3A4 inhibition	Increased CNI concentration leading to increased risk for side effects (e.g., renal impairment)
Direct oral anticoagulants (DOACs) Dabigatran Rivaroxaban	CCBs (diltiazem and verapamil) -via CYP-3A4 inhibition	Increased DOAC concentration leading to increased risk for side effects (e.g., bleeding)
HMG-CoA reductase inhibitors Atorvastatin Simvastatin	CCBs (diltiazem and verapamil) -via CYP-3A4 inhibition	Increased statin concentration leading to increased risk for side effects (e.g., myopathy)
Colchicine	CCBs (diltiazem and verapamil) -via CYP-3A4 inhibition	Increased colchicine concentration leading to increased risk for adverse effects (e.g., neuromuscular toxicity)
Elimination		
Lithium	Thiazide diuretics, Renin-angiotensin-aldosterone system inhibitors	Reduced clearance of lithium leading to increased risk of lithium toxicity
P-glycoprotein (P-gp)		
Dabigatran	Verapamil -via P-gp inhibition	Reduced P-gp efflux of dabigatran leading to increased levels and resulting increase in bleed risk
Digoxin	Verapamil, carvedilol -via P-gp inhibition	Reduced P-gp efflux of digoxin leading to increased levels and resulting increase risk of digoxin toxicity

cardiovascular outcomes [30], to very basic effects, like the combination of a potassium supplement with a thiazide diuretic to prevent the development of hypokalemia. Detrimental pharmacodynamic interactions can occur as well. For example, any medication that can increase blood pressure (e.g., NSAIDs, sympathomimetics, venlafaxine) can interact with antihypertensives by reducing their blood pressure lowering capability. In more specific examples, beta-blockers and non-dihydropyridine calcium channel blockers can interact with other classes of medications with negative inotropic effects (e.g., amiodarone and digoxin) leading to excessive bradycardia, and NSAID-induced sodium retention can antagonize the beneficial effects of diuretic therapy.

Table 3 Pharmacodynamic drug interactions with antihypertensives

Drug combinations		Clinical effect
Cautionary interactions		
Calcium Channel Blockers Diltiazem Verapamil	Beta-blockers	Atrioventricular block
Angiotensin-converting enzyme inhibitors	Angiotensin receptor blockers	Acute kidney injury Hyperkalemia
	Potassium-sparing diuretics Spironolactone Eplerenone Triamterene Amiloride	Hyperkalemia
	Sulfamethoxazole/trimethoprim	Hyperkalemia
	Potassium supplements	Hyperkalemia
	Non-steroidal anti-inflammatory drugs (NSAIDs) (e.g., ibuprofen, naproxen)	Acute kidney injury Hyperkalemia
Clonidine, methyldopa	CNS depressants (e.g., zolpidem, alprazolam)	Sedation
Clonidine	Non-cardio-selective beta-blockers (e.g., propranolol)	Unopposed alpha agonism upon beta-blocker withdrawal leading to hypertensive crisis
Antihypertensive (any)	NSAIDs	Reduced blood pressure lowering via sodium retention
	Sympathomimetic (e.g., pseudoephedrine, dextroamphetamine)	Reduced blood pressure lowering
	Venlafaxine	Reduced blood pressure lowering
Advantageous interactions		
Antihypertensive (any)	Antihypertensive (any)	Additive blood pressure lowering (but increased potential for symptomatic hypotension, orthostasis, syncope, etc.)
Renin-angiotensin-aldosterone system (RAAS) inhibitors	Diuretics	Balanced effects on serum potassium levels with diminished possibility for hypokalemia (with diuretic) or hyperkalemia (with RAAS inhibitor) Augmented blood pressure lowering given ability of RAAS inhibitors to suppress the activation of the RAAS typically seen as a result of diuretic therapy
Potassium binder	RAAS inhibitors	Lowers risk of hyperkalemia from the RAAS inhibitor
Potassium supplement	Diuretic	Lowers risk of hypokalemia from the diuretic

Applying knowledge of drug mechanisms can allow for purposeful selection of desirable pharmacodynamic interactions as well as careful avoidance of unwanted combinations. Table 3 includes a more comprehensive list of potential pharmacodynamic interactions involving antihypertensive medications.

Conclusions

Consideration of drug interactions is an important step when optimizing the use of antihypertensive agents and requires fundamental knowledge about the types of drug interactions and the pharmacologic effects of the agents involved. Although often more subtle than other high-profile drug-drug interactions, the potential for therapeutic antagonism or increased adverse effects remains important considerations when introducing antihypertensive drugs into existing complex medication regimens, or when non-antihypertensive drugs are initiated onto a background antihypertensive regimen. The need for ongoing pharmacovigilance in this area is great, given the substantial prevalence of antihypertensive use and the high rate of comorbidity and associated polypharmacy in this population.

Declarations

Conflict of Interest The authors declare no competing interests.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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•• Of major importance

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